

TrES-2b

R-0140

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_180609 · created 2026-07-16T07:32:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458278.73767 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1491 +/- 0.0064
Transit depth (Rp/Rs)^2	2.22 +/- 0.19 [%]
Orbital Inclination (inc)	85.29 +/- 0.56
Airmass coefficient 1 (a1)	0.964 +/- 0.01
Airmass coefficient 2 (a2)	0.03 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [444, 372]
Optimal Aperture	4.46
Optimal Annulus	4.64
Transit Duration (day)	0.0921 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1491 $\pm$ 0.0064 vs 0.1254 (deviation 3.7 σ)
Duration fit vs published	0.0921 d vs 0.0746 d (deviation 4.61 σ)
Mid-time O-C	-0.9 min
Depth SNR	23.3
Residual scatter	1.09 %

Light curve

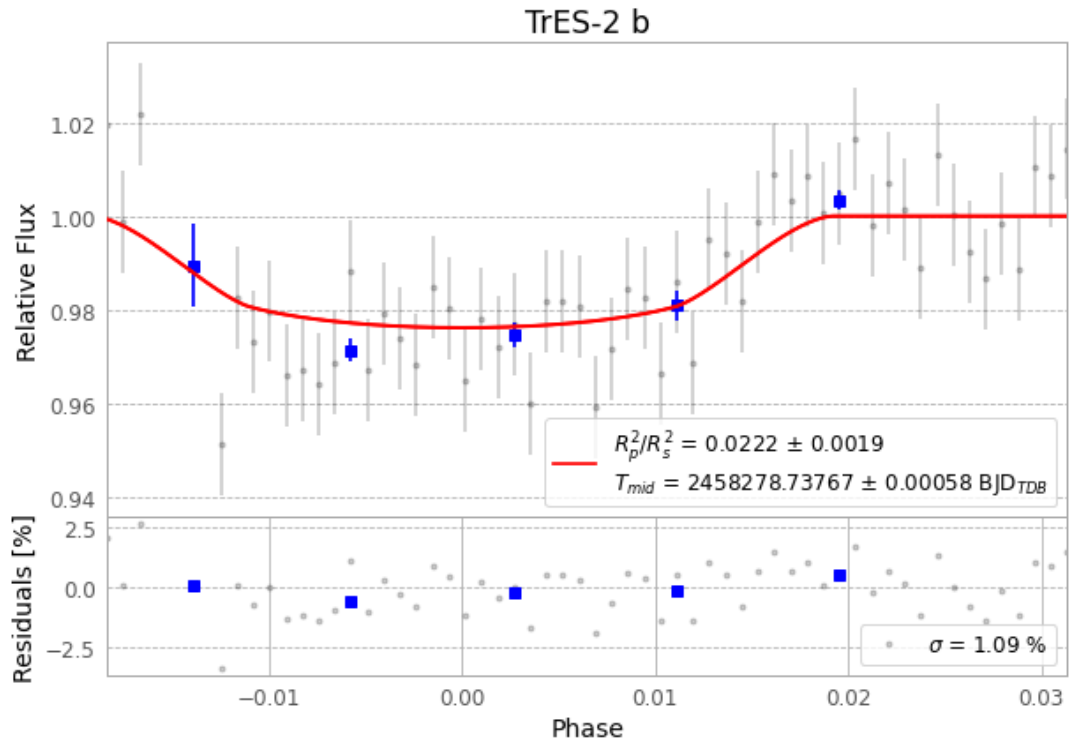


Figure 1 — FinalLightCurve_TrES-2 b_2018-06-08.png (SHA-256 f96311cb82b7e868...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0ac397eeb80d4dda...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

61 FITS frames (40 MB), TRES-2180609043012.FITS → TRES-2180609073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180609.log (fe2d8ebf8274...), FinalLightCurve_TrES-2 b_2018-06-08.png (f96311cb82b7...), FinalLightCurve_TrES-2 b_2018-06-08.csv (2bd11cf30b43...)

Compute resources

Wall time 260.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 07:32Z.